

Lucianu Dragoș - Adrian
 grupa 131
 evaluate ASC

1. $x = 125$

$y = 78$

a)
$$\begin{array}{l} 125 : 8 = 15 \text{ r } 5 \\ 15 : 8 = 1 \text{ r } 7 \\ 1 : 8 = 0 \text{ r } 1 \end{array} \quad \left| \begin{array}{l} \Rightarrow (125)_8 = \overline{175} = x \end{array} \right.$$

$$\begin{array}{l} 78 : 8 = 9 \text{ r } 6 \\ 9 : 8 = 1 \text{ r } 1 \\ 1 : 8 = 0 \text{ r } 1 \end{array} \quad \left| \begin{array}{l} \Rightarrow (78)_8 = \overline{116} = y \end{array} \right.$$

b)
$$\begin{array}{l} (\overline{1})_8 = (\overline{001})_2 \\ (\overline{7})_8 = (\overline{111})_2 \\ (\overline{5})_8 = (\overline{101})_2 \end{array} \quad \left| \begin{array}{l} \Rightarrow (x)_2 = \overline{001111101} = \overline{1111101} \end{array} \right.$$

$$\begin{array}{l} (\overline{1})_8 = (\overline{001})_2 \\ (\overline{1})_8 = (\overline{001})_2 \\ (\overline{6})_8 = (\overline{110})_2 \end{array} \quad \left| \begin{array}{l} \Rightarrow (y)_2 = \overline{001001110} = \overline{1001110} \end{array} \right.$$

c)
$$\begin{array}{l} \overline{0111} \quad \overline{1101} \\ (\overline{0111})_2 = (\overline{7})_{16} \\ (\overline{1101})_2 = (\overline{D})_{16} \end{array} \quad \left| \begin{array}{l} \Rightarrow (x)_{16} = \overline{7D} \end{array} \right.$$

$$\begin{array}{l} \overline{0100} \quad \overline{1110} \\ (\overline{0100})_2 = (\overline{4})_{16} \\ (\overline{1110})_2 = (\overline{E})_{16} \end{array} \quad \left| \begin{array}{l} \Rightarrow (y)_{16} = \overline{4E} \end{array} \right.$$

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$$c) (x+y)_8 = (\overline{175})_8 + (\overline{116})_8 = (313)_8$$

$$\begin{array}{r} 175 \\ 116 \\ \hline 313 \end{array}$$

$$(x-y)_{16} = (\overline{7D})_{16} - (\overline{4E})_{16} = (\overline{2F})_{16}$$

$$\begin{array}{r} 7D \\ - 4E \\ \hline 2F \end{array}$$

$$d) (\overline{313})_8 = 3 \cdot 8^2 + 1 \cdot 8^1 + 3 \cdot 8^0$$

$$= 192 + 8 + 3$$

$$= (203)_{10}$$

$$(\overline{2F})_{16} = 2 \cdot 16^1 + 15 \cdot 16^0$$

$$= 32 + 15$$

$$= (47)_{10}$$

$$e) (y)_2 = \overline{01001110}$$

complementul față de 1 al lui y este: $\overline{10110001}$

complementul față de 2 al lui y este:

$$\overline{10110001} + 1 = \overline{10110010}$$

$$z = x + y = \overline{10110010} + \overline{01111101}$$

Exercitiu 4: Conversia din Baze, Adunare
 pe 128

$$\begin{array}{r} 01111101 \\ 10110010 \\ \hline 10010111 \end{array}$$

$$z = x + y = \overline{00101111} \\ = \overline{1011111}$$

$$\begin{aligned} (z)_2 &= 1 \cdot 2^5 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \\ &= 32 + 8 + 4 + 2 + 1 \\ &= (47)_{10} \end{aligned}$$

$$4) \quad C = (\overline{1100})_2$$

$$2 = (\overline{0010})_2$$

$$y = (\overline{1001})_2$$

$$x = (\overline{1010})_2$$

$$0 = (\overline{0000})_2$$

regulă internă binară în format single:

$$\begin{array}{c|cccccccccccc} 1 & 100 & 0010 & 1 & 001 & 1010 & 0000 & 0000 & 0000 & 0000 \\ \hline & \text{caracteristica} & & & \text{fracție} & & & & & \end{array}$$

stim că în format single:

$$n = 32$$

$$h = 9 \rightarrow E_{\max} = \text{bias} = 2^{8-1} - 1 = 127$$

$$E_{\min} = -126$$

$$r = n - h = 32 - 9 = 23 \rightarrow |g| = 23$$

$$s = 1$$

$$\begin{aligned}
 e = \text{characteristica} &= 1 \cdot 2^7 + 1 \cdot 2^2 + 1 \cdot 2^0 \\
 &= 128 + 4 + 1 \\
 &= 133
 \end{aligned}$$

$$\begin{aligned}
 e = E + \text{bias} \Rightarrow E &= e - \text{bias} \\
 &= 133 - 127 \\
 &= 6 \rightarrow \text{---}
 \end{aligned}$$

~~E~~ test de overflow / underflow :

$$E \in [E_{\min}, E_{\max}]$$

$b \in [-126, 127] \Rightarrow$ format normalizat

$$g = g = \overline{0011010}$$

$$m = (-1)^g \cdot 2^E \cdot 1, \overline{g}$$

$$= (-1)^1 \cdot 2^6 \cdot 1, \overline{001101}$$

$$= (-1) \cdot \overline{1001101}$$

$$\Rightarrow \text{---} (-1) \cdot (2^6 \cdot 1, \overline{1})$$

$$\begin{aligned}
 (\overline{1001101})_2 &= 1 \cdot 2^6 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^0 \\
 &= 64 + 8 + 4 + 1
 \end{aligned}$$

$$= 77$$

$$\text{---} m = (-1) \cdot 77$$

$$m = (-1) \cdot 77$$

$$= -77$$

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2. $f: M_2^3 \rightarrow M_2^3$

$$f(x, y, z) = (f_1(x, y, z), f_2(x, y, z))$$

$$f_1(x, y, z) = \bar{x} \bar{z} + (\bar{y} \oplus z)$$

$$f_2(x, y, z) = (\bar{x} + y)(\bar{y} \oplus z)$$

index	x	y	z	$\bar{x} \bar{z}$	$\bar{y} \oplus z$	$\bar{x} + y$	f_1	f_2
0	0	0	0	1	1 ⊕	1	1	1
1	0	0	1	0	0 ⊕	1	0	0
2	0	1	0	1	0 ⊕	1	1	0
3	0	1	1	0	1 ⊕	1	1	1
4	1	0	0	0	1 ⊕	0	1	0
5	1	0	1	0	0 ⊕	0	0	0
6	1	1	0	0	0 ⊕	1	0	0
7	1	1	1	0	1 ⊕	1	1	1

FND:

$$f_1 = \bar{x} \cdot \bar{y} \cdot \bar{z} + \bar{x} \cdot \bar{y} \cdot z + \bar{x} \cdot y \cdot \bar{z} + \bar{x} \cdot y \cdot z + \cancel{\bar{x} \cdot \bar{y} \cdot \bar{z}} + \cancel{\bar{x} \cdot y \cdot z} + \bar{x} \cdot y \cdot z \quad (0, 2, 3, 4, 7)$$

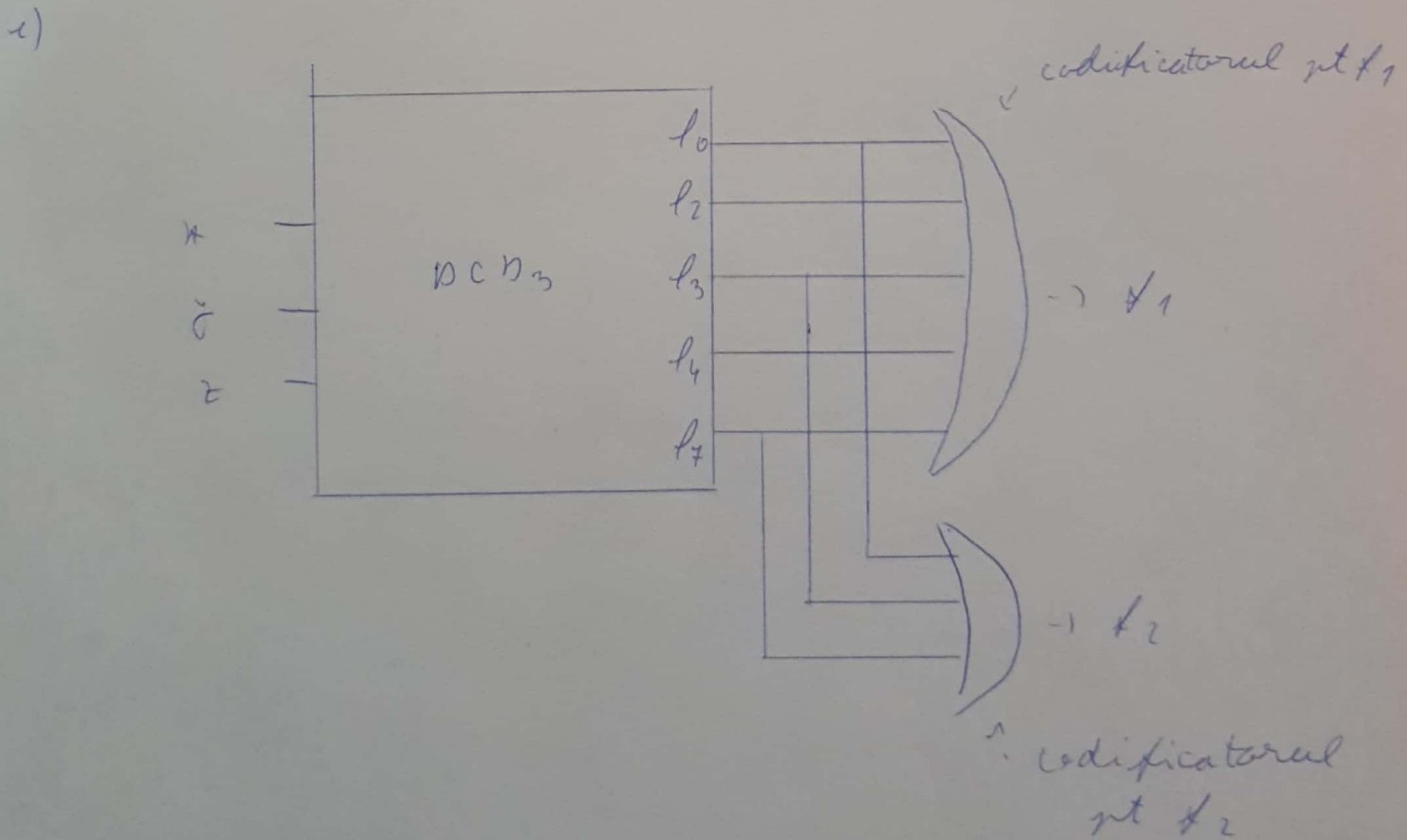
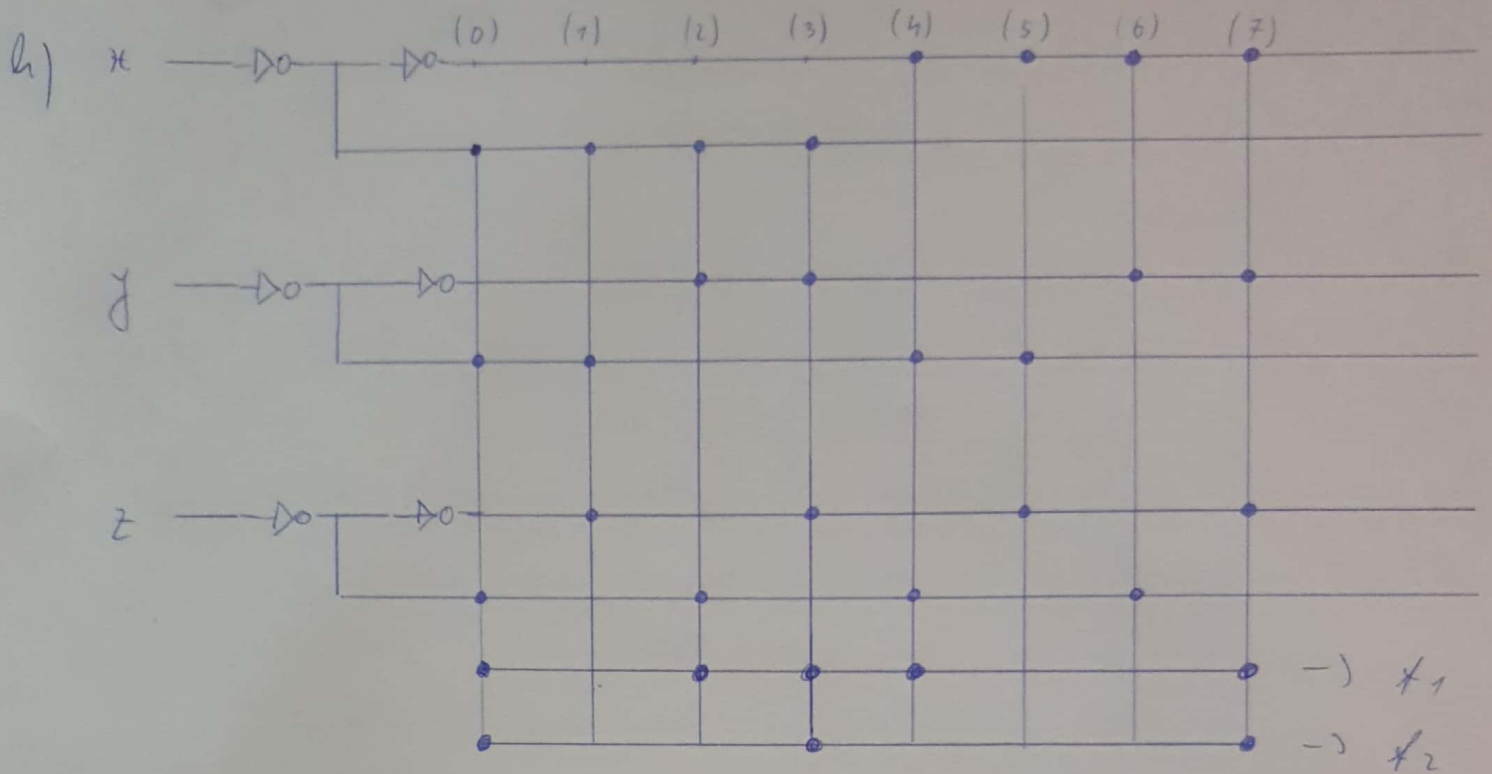
$$f_2 = \bar{x} \cdot \bar{y} \cdot \bar{z} + \bar{x} \cdot y \cdot \bar{z} + \bar{x} \cdot y \cdot z \quad (0, 3, 7)$$

FNC:

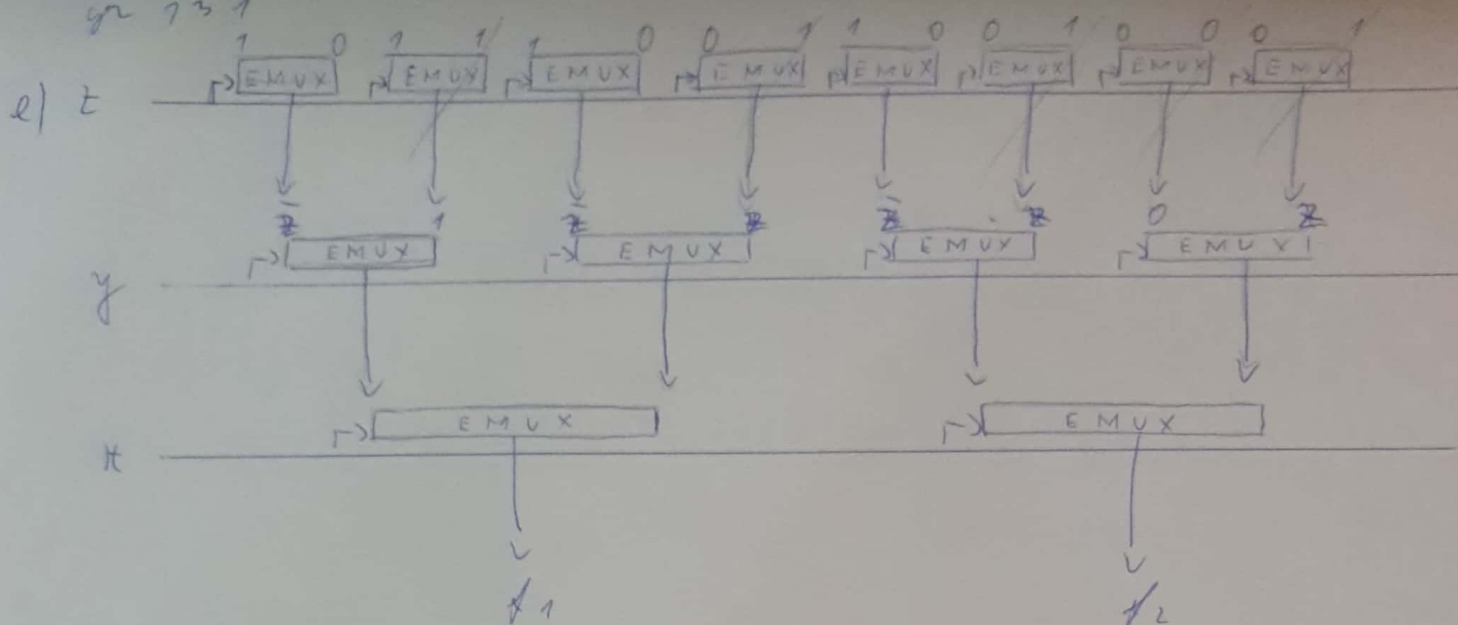
$$f_1 = (x + y + \bar{z}) \cdot (\bar{x} + y + \bar{z}) \cdot (\bar{x} + \bar{y} + z) \quad (1, 5, 6)$$

$$f_2 = (x + y + \bar{z}) \cdot (\bar{x} + \bar{y} + z) \cdot (\bar{x} + y + \bar{z}) \cdot (\bar{x} + y + z) \cdot (\bar{x} + \bar{y} + z) \quad (1, 2, 4, 5, 6)$$

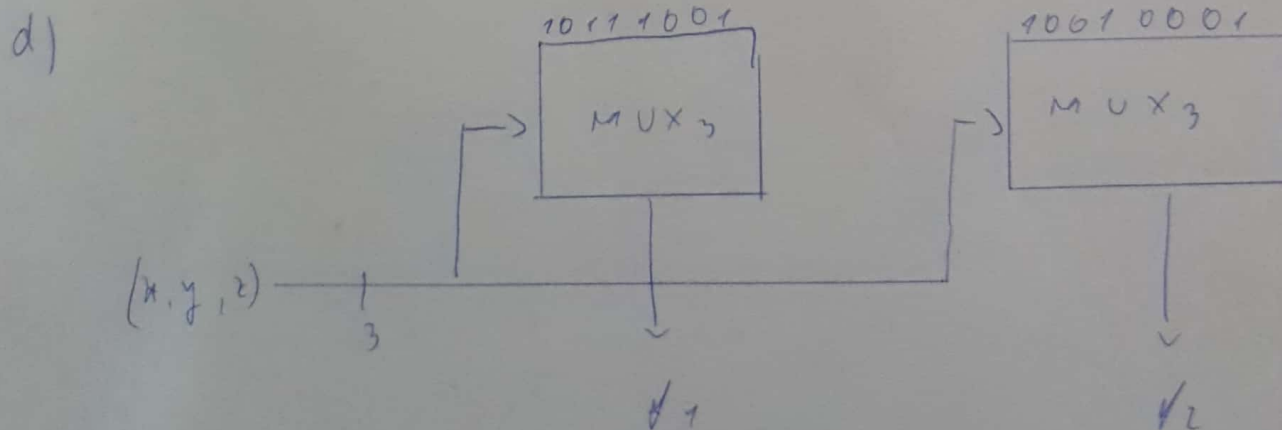
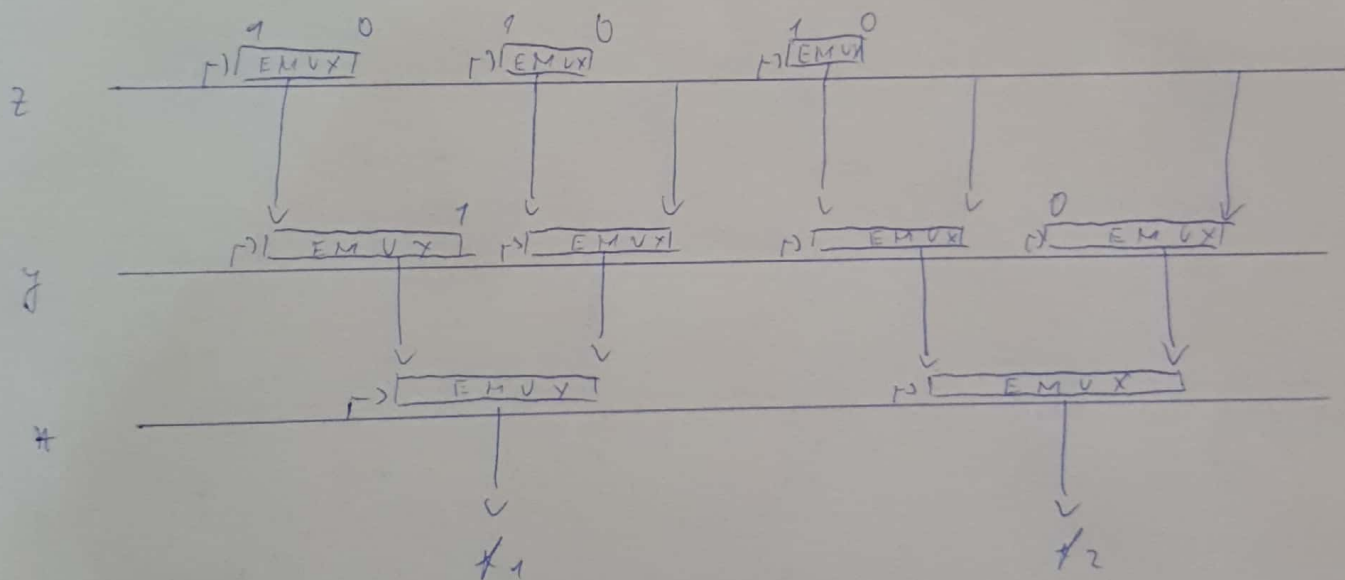
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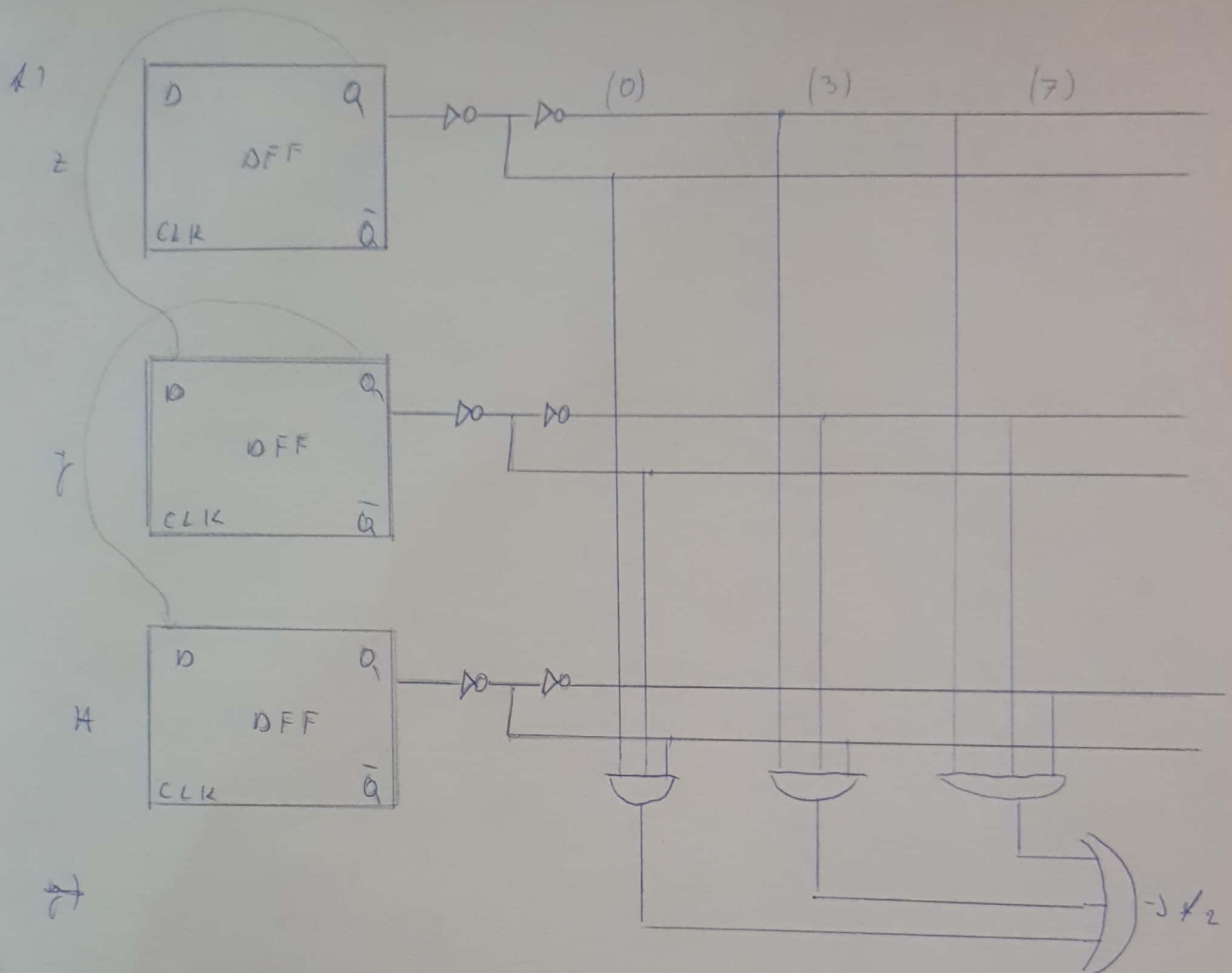
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reducere la maxim 30 multiglezari elementari



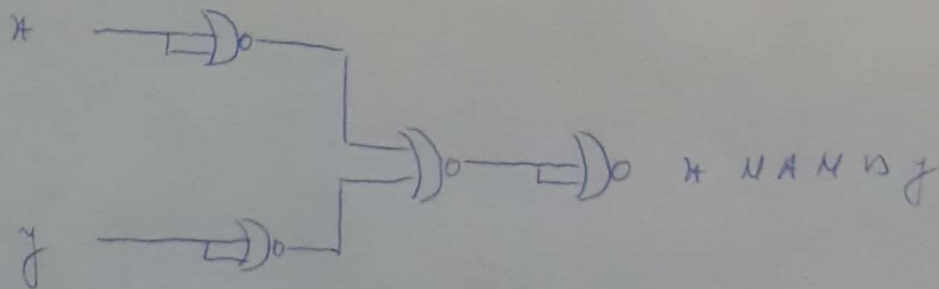
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g) A NAND y

$$\overline{A \cdot y} = \overline{(A \text{ NOR } A) \text{ NOR } (y \text{ NOR } y)}$$

$$= ((A \text{ NOR } A) \text{ NOR } (y \text{ NOR } y)) \text{ NOR } ((A \text{ NOR } A) \text{ NOR } (y \text{ NOR } y))$$



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3. a) nt let lt:

$$op = (\overline{000000})_2 = (00)_{16}$$

$$rs = (10)_2 = (\overline{01010})_2 = (0A)_{16}$$

$$rt = (10)_2 = (\overline{01011})_2 = (0B)_{16}$$

$$rd = (11)_2 = (\overline{01011})_2 = (0B)_{16}$$

$$shamt = (\overline{00000})_2 = (00)_{16}$$

$$funct = (\overline{101010})_2 = (2A)_{16}$$

~~bin: 0000 0010 1010 1110 1100~~

bin: 0000 0001 0100 1011 0101 1000 0010 1010

0x: 0 1 4 B 5 B 2 4

nt functiue beg =

$$op = 000100$$

$$rs = 01001$$

$$rt = 01011$$

$$imm = -4 = 1111 1111 1111 1100$$

repre imm comp data de 2

$$\begin{array}{r} 4 \text{ pe } 46 \text{ biti: } \begin{array}{cccc} 0000 & 0000 & 0000 & 0100 \\ \hline 1111 & 1111 & 1111 & 1011 \\ & & & 1 \\ \hline 1111 & 1111 & 1111 & 1100 \end{array} \end{array}$$

bin: 0001 0001 0010 1011 1111 1111 1111 1100

0x: 1 1 2 B F F F C

Exercise 2: Design - Address

1)

	1	4	5	7	8	ALU src	10	d	e	branch	Mem to Reg	ALU OP	ALU C+R2	REG WRITE	PC	#13
init	—	—	—	—	—	—	—	—	—	—	—	—	—	—	X	FFFFFFF (-1)
slt	X	010	FFFFF (-1)	1	?	1	X+4	X+4	0	0	1X	111	1	X+4	1	
beq	X+4	010	1	1	0	1	2	X-8	X-8	01	X	X1	110	0	X-8	1

2)

Instruction	Reg dst	ALU src	MEM TO REG	REG WRITE	MEM READ	MEM XWRITE	branch	Jump	ALU Op	ALU Op
—	0	1	1	1	1	0	0	0	0	1